

Hands-on Assignments for Using DDL Statements to Create and Manage Tables

Complete the below hands-on assignments before proceeding with the next Topic

1.

Create table dept1 as below with following columns: DEPTNO INTEGER PK,DNAME VARCHAR2(30) NOT NULL,LOC VARCHAR2(30) NOT NULL

SQL> create table dept1(deptno integer primary key,dname varchar2(30) not null, loc varchar2(30) not null);

```
SQL> create table dept1(deptno integer primary key,dname varchar2(30) not null,loc varchar2(30) not null);
Table created.

SQL> desc dept1;
Name                               Null?    Type
-----
DEPTNO                             NOT NULL NUMBER(38)
DNAME                              NOT NULL VARCHAR2(30)
LOC                                 NOT NULL VARCHAR2(30)
```

2.

Create table emp1 with the following columns EMPNO INTEGER PK,ENAME VARCHAR2(20) NOT NULL,SAL NUMBER(10,2) check(sal>5000),mgr number FK(EMPNO),DEPTNO INTEGER FK(DEPTNO)

SQL> create table emp1(empno integer primary key,ename varchar2(20) not null,sal number(10,2) check (sal>5000),mgr number,deptno integer, foreign key(deptno) references dept1(deptno));

```
SQL> create table emp1(empno integer primary key,ename varchar2(20) not null,sal number(10,2) check(sal>5000),mgr number,deptno integer,foreign key(deptno) references dept1(deptno));
Table created.

SQL> desc emp1;
Name                               Null?    Type
-----
EMPNO                             NOT NULL NUMBER(38)
ENAME                              NOT NULL VARCHAR2(20)
SAL                                NUMBER(10,2)
MGR                                NUMBER
DEPTNO                             NUMBER(38)
```

3.

Create tables Dept11,Emp11 from Dept1,Emp1.

SQL> create table dept11 as select * from dept1;

```
SQL> create table dept11 as select * from dept1;
```

Table created.

```
SQL> desc dept11;
```

Name	Null?	Type
DEPTNO		NUMBER(38)
DNAME	NOT NULL	VARCHAR2(30)
LOC	NOT NULL	VARCHAR2(30)

```
SQL> create table emp11 as select * from emp1;
```

```
SQL> create table emp11 as select * from emp1;
```

Table created.

```
SQL> desc emp11;
```

Name	Null?	Type
EMPNO		NUMBER(38)
ENAME	NOT NULL	VARCHAR2(20)
SAL		NUMBER(10,2)
MGR		NUMBER
DEPTNO		NUMBER(38)

4.

Add a new column Address-Varchar2(30) in the Emp1 table.

```
SQL> alter table emp1 add Address varchar2(30);
```

```
SQL> alter table emp1 add Address varchar2(30);
```

Table altered.

```
SQL> desc emp1;
```

Name	Null?	Type
EMPNO	NOT NULL	NUMBER(38)
ENAME	NOT NULL	VARCHAR2(20)
SAL		NUMBER(10,2)
MGR		NUMBER
DEPTNO		NUMBER(38)
ADDRESS		VARCHAR2(30)

5.

Rename SAL column in Emp1 table to SALARY.

```
SQL> alter table emp1 rename column sal to salary;
```

```
SQL> alter table emp1 rename column sal to salary;
```

Table altered.

```
SQL> desc emp1;
```

Name	Null?	Type
EMPNO	NOT NULL	NUMBER(38)
ENAME	NOT NULL	VARCHAR2(20)
SALARY		NUMBER(10,2)
MGR		NUMBER
DEPTNO		NUMBER(38)
ADDRESS		VARCHAR2(30)

6.

Rename the FK constraint names and verify the names for the EMP1.

7.

Modify ENAME Column by increasing the length of the column to 40.

```
SQL> alter table emp1 modify ename varchar2(40);
```

```
SQL> alter table emp1 modify ename varchar2(40);
```

Table altered.

```
SQL> desc emp1;
```

Name	Null?	Type
EMPNO	NOT NULL	NUMBER(38)
ENAME	NOT NULL	VARCHAR2(40)
SALARY		NUMBER(10,2)
MGR		NUMBER
DEPTNO		NUMBER(38)
ADDRESS		VARCHAR2(30)

8.

Drop the NOT NULL constraint for ENAME.

```
SQL> ALTER TABLE emp1 MODIFY ename varchar2(40) NULL;
```

```
SQL> ALTER TABLE emp1 MODIFY ename varchar2(40) NULL;
```

Table altered.

```
SQL> desc emp1;
```

Name	Null?	Type
EMPNO	NOT NULL	NUMBER(38)
ENAME		VARCHAR2(40)
SALARY		NUMBER(10,2)
MGR		NUMBER
DEPTNO		NUMBER(38)
ADDRESS		VARCHAR2(30)

```
SQL>
```

9.

Create comment on Dept1 table as 'Depts of WIPRO'

```
SQL> comment on table dept1 is 'Depts of WIPRO';
```

```
SQL> comment on table dept1 is 'Depts of WIPRO';  
Comment created.
```

10.

Create comment on Deptno of Dept1 as 'Deptno of WIPRO'

```
SQL> COMMENT ON COLUMN dept1.deptno is 'deptno of WIPRO';
```

```
SQL> COMMENT ON COLUMN dept1.deptno is 'deptno of WIPRO';  
Comment created.
```

11.

Create a comment on Emp1 table as 'Employees of WIPRO'

```
SQL> comment on table emp1 is 'Employees of WIPRO';
```

```
SQL> comment on table emp1 is 'Employees of WIPRO';  
Comment created.
```

12.

Create a comment on Empno of Emp1 table as 'Empno of WIPRO'

```
SQL> COMMENT ON COLUMN emp11.empno is 'empno of WIPRO';
```

```
SQL> COMMENT ON COLUMN emp11.empno is 'empno of WIPRO';
Comment created.
```

13.

Remove all the comment on the tables and columns.

```
SQL> COMMENT ON table emp1 IS '';
```

```
SQL> COMMENT ON COLUMN emp1.empno IS '';
```

```
SQL> COMMENT ON table dept1 IS '';
```

```
SQL> COMMENT ON COLUMN dept1.deptno IS '';
```

```
SQL> COMMENT ON table emp1 IS '';
Comment created.

SQL> COMMENT ON COLUMN emp1.empno IS '';
Comment created.

SQL> COMMENT ON table dept1 IS '';
Comment created.

SQL> COMMENT ON COLUMN dept1.deptno IS '';
Comment created.
```

14.

Set sal,ename columns of Emp1 table as unused.

```
ALTER TABLE emp1 SET UNUSED (salary,ename);
```

```
SQL> ALTER TABLE emp1 SET UNUSED (salary,ename);
Table altered.

SQL> desc emp1;
Name                               Null?    Type
-----
EMPNO                               NOT NULL NUMBER(38)
MGR                                  NUMBER
DEPTNO                              NUMBER(38)
ADDRESS                             VARCHAR2(30)

SQL>
```

15.

Drop the unused columns.

```
ALTER TABLE emp1 DROP UNUSED COLUMNS;
```

```
SQL> alter table emp1 drop UNUSED COLUMNS;  
Table altered.
```

16.

Drop Emp1,Dept1 Tables.

```
SQL> drop table emp1;
```

Table dropped. SQL>

```
drop table dept1; Table
```

dropped.

```
SQL> drop table emp1;  
Table dropped.  
SQL> drop table dept1;  
Table dropped.  
SQL> drop table emp1;  
drop table emp1  
*  
ERROR at line 1:  
ORA-00942: table or view does not exist  
  
SQL> desc dept1;  
ERROR:  
ORA-04043: object dept1 does not exist
```

17.

Create Table Emp1 from EMP table by copying data also. Check your data.

```
SQL> create table emp1 as select * from emp; Table
```

created.

```
SQL> create table emp1 as select * from emp;
```

Table created.

```
SQL> select * from emp1;
```

EMPID	ENAME	JOB	MGR	HIREDATE	SAL	COMM
7839 10	KING	PRESIDENT		17-NOV-81	5000	
7788 30	BLAKE	MANAGER	7839	01-MAY-81	2850	
7799 30	Johnson	MANAGER	7342	01-JUN-87	2855	

18.

Rename Emp1 to EMP_Test.

```
SQL> alter table emp1 rename to emp_test; Table
```

altered.

```
SQL> alter table emp1 rename to emp_test;
```

Table altered.

```
SQL> select * from emp_test;
```

EMPID	ENAME	JOB	MGR	HIREDATE	SAL	COMM
7839 10	KING	PRESIDENT		17-NOV-81	5000	
7788 30	BLAKE	MANAGER	7839	01-MAY-81	2850	
7799 30	Johnson	MANAGER	7342	01-JUN-87	2855	

19.

Truncate Emp_Test table and confirm your delete.

```
SQL> truncate table emp_test;

Table truncated.

SQL> select * from emp_test;

no rows selected
```

20.

Create Emp2 from Emp by only copying Empno,Ename,sal columns by copying Data.

SQL> create table emp2 as select empid,ename,sal from emp;

```
SQL> create table emp2 as select empid,ename,sal from emp;

Table created.

SQL> desc emp2;
Name                               Null?    Type
-----
EMPID                               NUMBER(4)
ENAME                               VARCHAR2(10)
SAL                                 NUMBER(7,2)
```

21.

Drop Emp2 Table.

SQL> drop table emp2; Table

dropped.

```
SQL> drop table emp2;

Table dropped.

SQL> desc emp2;
ERROR:
ORA-04043: object emp2 does not exist
```

22.

Create Emp2 table again from EMP table without copying data.

SQL> CREATE TABLE emp2 AS SELECT * FROM emp WHERE 1=0;

Table created.

```
SQL> CREATE TABLE emp2 AS SELECT * FROM emp WHERE 1=0;
```

Table created.

```
SQL> desc emp2;
```

Name	Null?	Type
EMPID		NUMBER(4)
ENAME		VARCHAR2(10)
JOB		VARCHAR2(9)
MGR		NUMBER(4)
HIREDATE		DATE
SAL		NUMBER(7,2)
COMM		NUMBER(7,2)
DEPTID		NUMBER(2)

```
SQL> select * from emp2;
```

no rows selected

23.

Drop Emp2 table.

```
SQL> drop table emp2; Table
```

dropped.

```
SQL> select * from emp2;
```

no rows selected

24.

Flashback the first Emp2 table that contained data and check your table.

```
SQL> FLASHBACK TABLE emp2 TO BEFORE DROP;
```

```
SQL> SHOW RECYCLEBIN;
ORIGINAL NAME      RECYCLEBIN NAME      OBJECT TYPE  DROP TIME
-----
DEPT               BIN$pTu+QthxRDurrePEUZZTWA==$0  TABLE      2024-09-12:10:21:56
DEPT1              BIN$YvNpjWiZS4Gu3dykRlS+/Q==$0  TABLE      2024-09-28:14:25:04
DEPT1              BIN$Aj02V40eRC+/rPLFTKIaoQ==$0  TABLE      2024-09-21:11:49:11
DEPT11             BIN$17Wv73TcRZWYb0rhj85+1A==$0  TABLE      2024-09-21:11:59:53
EMP                BIN$oJI9pKsFQC6EOdPMIvWrLA==$0  TABLE      2024-09-12:10:21:49
EMP1               BIN$Dxs3bh7vSoOUvq+EmjQ1Nw==$0  TABLE      2024-09-28:14:31:22
EMP1               BIN$b5ai/y0bQ5ScvsrR7oVkoQ==$0  TABLE      2024-09-28:14:24:57
EMP2               BIN$DwsV7foFQAqQpx3UFbODLA==$0  TABLE      2024-09-30:11:17:35
EMP2               BIN$9dyGPwVPSg2LvVDuZTHzHA==$0  TABLE      2024-09-30:11:06:21
EMP2               BIN$fhw62iV8Sy2SapAZK+EPQQ==$0  TABLE      2024-09-30:10:57:21
SQL> FLASHBACK TABLE emp2 TO BEFORE DROP;
```

Flashback complete.

```
SQL> select * from emp2;
```

EMPID	ENAME	JOB	MGR	HIREDATE	SAL	COMM
7839	KING	PRESIDENT		17-NOV-81	5000	
10						
7788	BLAKE	MANAGER	7839	01-MAY-81	2850	
30						

25.

Flashback the recently dropped table by renaming to EMP2_1.

```
SQL> FLASHBACK TABLE emp2 TO BEFORE DROP RENAME TO emp2_1;
```

Flashback complete.

```
SQL> FLASHBACK TABLE emp2 TO BEFORE DROP RENAME TO emp2_1;
Flashback complete.
SQL> select * from emp2_1;
```

EMPID	ENAME	JOB	MGR	HIREDATE	SAL	COMM
7839	KING	PRESIDENT		17-NOV-81	5000	
10						
7788	BLAKE	MANAGER	7839	01-MAY-81	2850	
30						
7799	Johnson	MANAGER	7342	01-JUN-87	2855	
30						

26.

Check your tables.

TABLE_NAME

REGIONS

LOCATIONS

DEPARTMENTS

JOBS

EMPLOYEES

JOB_HISTORY

DEPT

EMP

LODGE_DETAILS

SKILL_DETAILS

EMP_DETAILS

TABLE_NAME

EMP_SKILL

DEPT11

EMP11

STUDENT

SALGRADE

DEPT_1

EMP_1

PRODUCTSTB

DEPT1

EMP2_1

COUNTRIES

TABLE_NAME

EMP_TEST

23 rows selected.